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2.2P

Counter.cs

```
namespace CounterTask
{
    public class Counter
    {
        private int _count;

        private string _name;
        public Counter(string name)
        {
            _name = name;
            _count = 0;
        }

        public void Increment()
        {
            _count++;
        }

        public void Reset()
        {
            _count = 0;
        }

        public string Name
        {
            get
            {
                return _name;
            }
            set
            {
                _name = value;
            }
        }

        public int Ticks
        {
            get
            {
                return _count;
            }
        }

        public void ResetByDefault()
        {
            _count = (int)2147483647912;
        }
    }
}
```

```

//Answer to Q13
//No, the code does not run — it throws a compile-time error.
//The value 2147483647912 exceeds the maximum limit of the int data type in C#,
// which is 2,147,483,647. To store larger values,
// we must use a long data type or cast it with overflow handling (e.g., unchecked).
//To fix this , we can use a long data type or handle the overflow.
// The 'unchecked' block tells the compiler to ignore overflow errors,
// so the large number wraps around and is stored as a negative int.}
    }
}
}

```

Program.cs

```

namespace CounterTask
{
    internal class Program
    {
        private static void PrintCounters(Counter[] counters)
        {
            foreach (Counter c in counters)
            {
                Console.WriteLine("{0} is {1}", c.Name, c.Ticks);
            }
        }
        public static void Main()
        {
            Counter[] myCounters = new Counter[3];
            myCounters[0] = new Counter("Counter 1");
            myCounters[1] = new Counter("Counter 2");
            myCounters[2] = myCounters[1];
            for (int i = 1; i <= 9; i++)
            {
                myCounters[0].Increment();
            }
            for (int i = 1; i <= 14; i++)
            {
                myCounters[1].Increment();
            }
            Program.PrintCounters(myCounters);
            myCounters[2].Reset();
            Program.PrintCounters(myCounters);
        }
    }
}

```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
● ~/Desktop/00PAssignments/P22/CounterTask > dotnet run
Counter 1 is 9
Counter 2 is 14
Counter 2 is 14
Counter 1 is 9
Counter 2 is 0
Counter 2 is 0
○ ~/Desktop/00PAssignments/P22/CounterTask > █
```